1. Identify the mathematical aspects of a real-world problem situation including identifying the significant variables (Point 6 in the list)

The first step in solving this problem is to identify all the necessary variables related to calculating the overall fault rate and understanding their meaning in a real-world context.

- **Specific Analysis:** Students need to extract six key numerical values from the two tables: Number of Electrix video players (2000), number of Tronics video players (7000), number of Electrix audio players (6000), number of Tronics audio players (1000), Electrix video fault rate (5%), Tronics video fault rate (4%), Electrix audio fault rate (3%), and Tronics audio fault rate (2%). This requires students to transform unstructured table data into structured variables suitable for mathematical analysis.

2. Translate a problem into a standard mathematical representation or algorithm (Point 9 in the list)

The core challenge of this question lies in the fact that the tables provide fault rates for each product **category**, not the **overall** fault rate. Students need to realize they cannot simply take the arithmetic average of the two companies' fault rate percentages (e.g., $(5\% + 3\%)/2$) because the production volumes for the two categories are different; instead, they must use a **weighted average** or calculate the total number of faults.

- **Specific Analysis:** Students need to design a multi-step calculation method:
  1. Calculate the number of faults for each type of player for each company (e.g., Electrix video faults = $2000 \times 5\%$).
  2. Calculate the total number of faults for each company.
  3. Calculate the total production volume for each company.
  4. Calculate the overall fault rate percentage for each company (total faults divided by total production).
     This is the process of translating a real-world problem into a standard mathematical model (weighted average).

3. Construct and communicate explanations and arguments in the context of the problem (Point 7 in the list)

The question explicitly asks to "Show your calculations using the data in the tables above".

- **Specific Analysis:** Students must not only arrive at the final answer (which company is lower) but also provide clear, organized calculation steps and comparison processes to support their conclusion. This demonstrates the ability for mathematical argumentation and communication, using mathematical language and calculation results to prove a point.

1. 识别现实世界问题情境的数学方面，包括识别重要变量 (列表中的第 6 点)

解决这个问题的第一步是识别与计算总体故障率相关的所有必要变量，并理解它们在现实情境中的含义 (p. 1)。

- **具体分析：** 学生需要从两个表格中提取六个关键数值：Electrix 视频播放器数量 (2000)、Tronics 视频播放器数量 (7000)、Electrix 音频播放器数量 (6000)、Tronics 音频播放器数量 (1000)、Electrix 视频故障率 (5%)、Tronics 视频故障率 (4%)、Electrix 音频故障率 (3%) 和 Tronics 音频故障率 (2%) (p. 1)。这要求学生将非结构化的表格数据转化为可用于数学分析的结构化变量。

2. 将问题转化为标准的数学表示或算法 (列表中的第 9 点)

此题的核心挑战在于，表格中提供了的是每个产品**类别**的故障率，而不是**总体**故障率。学生需要意识到不能简单地将两个公司的故障率百分比（例如 (5% + 3%) / 2）进行算术平均，因为两类产品的产量不同，而是必须使用**加权平均数**或计算总的故障数量。

- **具体分析：** 学生需要设计一个涉及多个步骤的计算方法：
  1. 计算每家公司每种类型播放器的故障数量（例如 Electrix 视频故障数 = $2000 \times 5\%$）。
  2. 计算每家公司的总故障数量。
  3. 计算每家公司的总产量。
  4. 计算每家公司的总体故障率百分比（总故障数除以总产量）。
     这是一个将现实问题转化为标准数学模型（加权平均）的过程。

3. 构建并传达基于问题背景的解释和论证 (列表中的第 7 点)

题目明确要求“Show your calculations using the data in the tables above”（展示你使用上述表格中数据的计算过程）(p. 1)。

- **具体分析：** 学生必须不仅得出最终答案（哪家公司更低），还要提供清晰、有条理的计算步骤和比较过程，以支持他们的结论。这体现了数学论证和交流的能力，即用数学语言和计算结果来证明一个观点。